Metal Package PMT

Photosensor Modules H11706 Series

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The H11706 series is an optical sensor module containing a metal package photomultiplier tube, a high-voltage power supply circuit, and a shutter circuit. The shutter circuit electrically protects the photomultiplier tube from excessive light such as excitation light that illuminates samples.
The H11706 series offers a choice of photomultiplier tubes with different photocathodes: multialkali photocathodes with sensitivity extending to the near infrared region or enhanced red sensitivity and GaAsP photocathodes with high sensitivity in the visible region. The H11706 series also includes “P type” with high gain and low dark count specifically selected for photon counting applications.

Product Variations

Parameter	Spectral Response	Features
H11706-01	230 nm to 870 nm	For visible to near IR range
H11706-20	230 nm to 920 nm	Infrared-extended multialkali photocathode with enhanced sensitivity
H11706-40	300 nm to 720 nm	GaAsP photocathode, QE 40 % at peak wavelength
H11706P-01	230 nm to 870 nm	For visible to near IR range, For photon counting
H11706P-40	300 nm to 720 nm	GaAsP photocathode, For photon counting

This product can't be used at vacuum environment or reduced pressure environment.

Specifications

(at +25 °C)

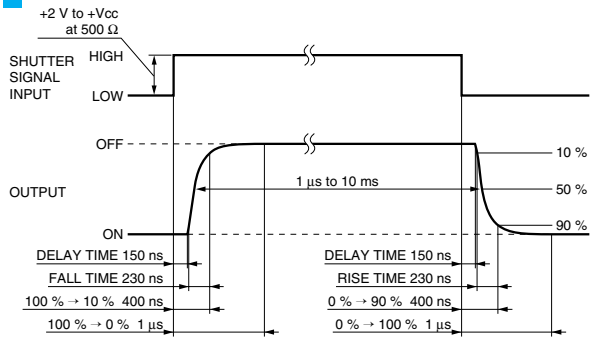
Parameter				H11706-01	H11706-20	H11706-40	Unit
Input Voltage				+14.5 to +15.5			V
Max. Input Voltage				+16			V
Max. Input Current *1*2				10			mA
Max. Average Output Signal Current*2				100		2	μA
Max. Control Voltage				+1.1		+0.9	V
Recommended Control Voltage Adjustment Range				+0.5 to +1.1		+0.5 to +0.8	V
Control Voltage Input Impedance				30			kΩ
Effective Area				φ8		φ5	mm
Peak Sensitivity Wavelength				400	630	580	nm
Cathode	Luminous Sensitivity		Min.	100	350	—	μA/lm
			Typ.	200	500	—	
	Red / White Ratio		Typ.	0.2	0.45	—	—
	Radiant Sensitivity		Typ.	77 (400 nm)	78 (630 nm)	176 (550 nm)	mA/W
Anode	Standard Type	Luminous Sensitivity *2	Min.	100	350	—	A/lm
			Typ.	400	1000	—	
		Radiant Sensitivity *2	Typ.	1.5×10^5 (400 nm)	1.5×10^5 (630 nm)	8.8×10^4 (550 nm)	A/W
		Dark Current *2*3	Typ.	1	10	5	nA
			Max.	10	100	—	
	P Type	Radiant Sensitivity *2	Typ.	1.5×10^5 (400 nm)	—	17.6×10^4 (550 nm)	A/W
		Dark Count *2*3	Typ.	600	—	6000	s^{-1}
			Max.	1000	—	18000	
Rise Time *2				0.57		1	ns
Ripple Noise *2*4 (peak to peak)			Max.	0.8			mV
Settling Time			Max.	0.2 *5		0.2 *6	s
Operating Ambient Temperature *7				+5 to +50		+5 to +35	°C
Storage Temperature *7				-20 to +50			°C
Weight				125		135	g

*1: At +15 V input voltage, output current equal to dark current, and without shutter signal input
*2: H11706-01/-20: Control voltage = +1.0 V, H11706-40: Control voltage = +0.8 V
*3: After 30 minutes storage in darkness, output of dark current.
*4: Cable RG-174/U, Cable length 450 mm, Load resistance = 1 MΩ, Load capacitance = 22 pF
*5: The time required for the output to reach a stable level following a change in the control voltage from +1.0 V to +0.5 V.
*6: The time required for the output to reach a stable level following a change in the control voltage from +0.9 V to +0.5 V.
*7: No condensation

(at +25 °C)

Parameter			Description / Value	Unit
Shutter Mode	Mode		Nomally ON	—
	Shutter Width (FWHM)		1 µs to 10 ms	—
	Rise Time	Typ.	230	ns
	Fall Time	Typ.	230	ns
	Repetition Rate	Max.	10	kHz
	Switching Ratio	Typ.	10^3	—
	Delay Time	Typ.	150	ns
Shutter Signal	Input LOW Level	Min.	0	V
		Max.	+0.4	
	Input HIGH Level	Min.	+2.0	V
		Max.	Vcc	
	Input Impedance		500	Ω

Shutter Characteristics



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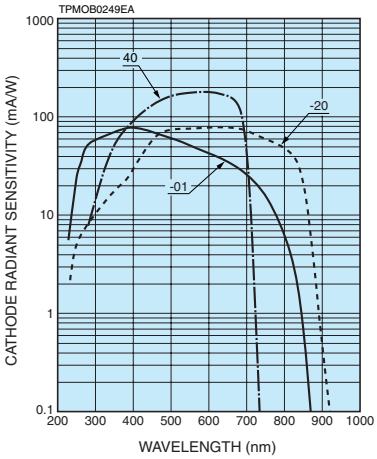
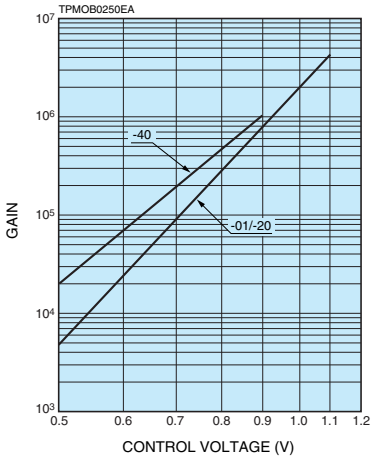
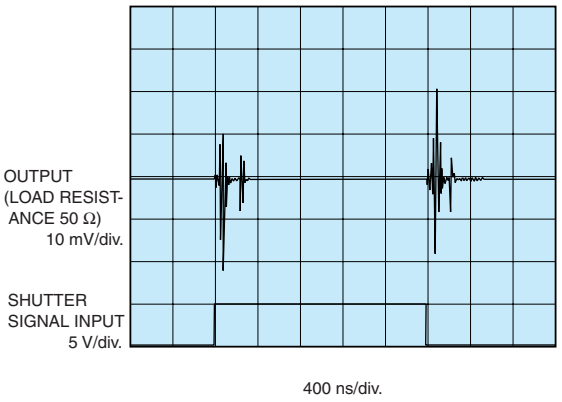
Characteristics (Cathode radiant sensitivity, Gain)



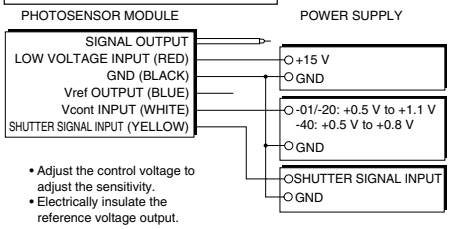


Switching Noise



Sensitivity Adjustment Method

VOLTAGE PROGRAMMING



- Adjust the control voltage to adjust the sensitivity.
- Electrically insulate the reference voltage output.

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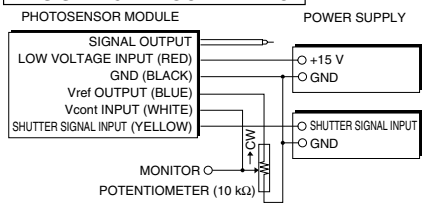
RESISTANCE PROGRAMMING



* When using a potentiometer to adjust sensitivity, monitor the control voltage so it does not exceed +1.1 V (-01/-20) / +0.9 V (-40).

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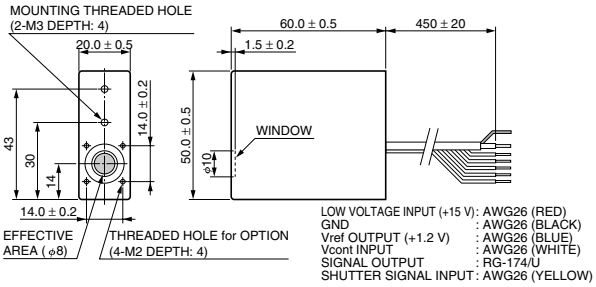
Dimensional Outlines (Unit: mm)

●H11706-01/-20



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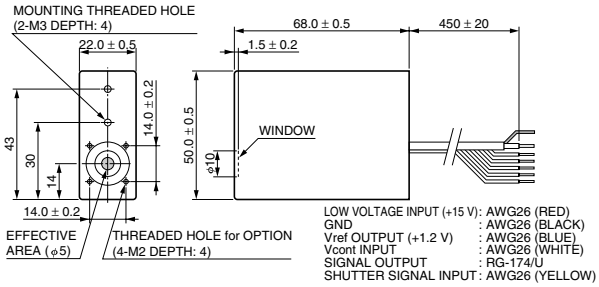
●H11706-40



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